

Libby Chambers

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Education

UNIVERSITY OF MICHIGAN

Bachelor of Science in Biomedical Engineering
GPA: 3.5/4.0

Ann Arbor, MI
December 2025

Research Experience

UNIVERSITY

Research Assistant
PI: Dr. Brendon Baker

Somewhere
April 2026 - Current

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UNIVERSITY OF MICHIGAN BIOMEDICAL ENGINEERING DEPARTMENT

Capstone Project
Instructors: Dr.

Ann Arbor, MI
Sept. 2025 - Dec. 2025

- Designed and built a functional electronic biomedical monitoring device for epilepsy detection
- Assembled circuit components by soldering electrical components onto protoboards for rapid prototyping
- Researched and applied ISO and ASTM methodologies in the reaction of the device
- Collected experimental data and iteratively improved device functionality through troubleshooting

KRESGE HEARING RESEARCH INSTITUTE

Research Assistant
PI: Dr. Renee Banakis-Hartl

Ann Arbor, MI
Jan. 2024 - Aug. 2025

- Research

JOHANNES GUTENBERG UNIVERSITY

Research Intern
PI: Dr. Kristina Endres

Mainz, Germany
June 2024 - Aug. 2024

- Conducted laboratory research on neurodegenerative disease using organoid and tissue models
- Performed microscopy-based imaging and sample preparation for biological analysis
- Maintained detailed lab notebooks and electronic records in compliance with research standards
- Developed a Python-based image analysis pipeline to automate organoid quantification
- Collaborated with researchers to troubleshoot experiments and optimize workflows

Research Conferences and Publications

ASSOCIATION FOR RESEARCH IN OTOLARYNGOLOGY MIDWINTER MEETING

Poster Presenter

“Impact of Cochlear Implants For Single-Sided Deafness on Head Movement and Resulting of Interaural Level and Timing Difference Cues During a Novel Localization Task”

Orlando, FL
Feb. 2025

UNIVERSITY OF MICHIGAN NEUROSCIENCE HACKATHON

Participant and Presenter
“Expert Disagreements of EEG Recordings”

Ann Arbor, MI
Aug. 2025

RHINE MAIN NEUROSCIENCE NETWORK BIENNIAL MEETING

Oberwiesel, Germany
June 2024

Relevant Coursework

TISSUE ENGINEERING

Aug. 2025 - Dec. 2025

Instructor: Dr. Putnam

- Performed hands-on laboratory work including media preparation, cell culture and sterile technique
- Maintained laboratory cleanliness and followed GLP-like protocols
- Recorded experimental observations and analyzed data to evaluate system performances

ARTIFICIAL INTELLIGENCE FOR BME

Aug. 2025 - Dec. 2025

Instructor: Dr. Chandrasekaran

- Built disease modeling frameworks using Python and MATLAB to analyze biomedical datasets
- Applied machine learning techniques for classification and prediction in biological systems

BIOREACTION ENGINEERING AND DESIGN

Jan. 2025 - May 2025

Instructor: Dr. Shea

- Biochemical pathway engineering, mass and energy balance, cell growth and differentiation, cell engineering, bioreactor design, and analysis of the human body, organs, tissues, and cells as bioreactors

BIOLOGY LABORATORIES (BME 200S)

Instructor:

- Conducted PCR and gel electrophoresis to characterize microbial DNA from gut microbiome samples
- Evaluated how dietary fiber types influence microbial composition and short-chain fatty acid production
- Designed and interpreted experiments linking microbial metabolism to host health outcomes

Teaching Experience

WASHTENAW PUBLIC SCHOOLS

Math, Science, and English Instructor, Substitute Teaching

Paraprofessional Instructor, Substitute Teaching

Ann Arbor, MI

Jan. 2026 - April 2026

Jan. 2026 - April 2026

MAIZE AND BLUE TUTORS

Tutor

Ann Arbor, MI

June 2025 - Aug. 2025

UNIVERSITY OF MICHIGAN

Organic Chemistry Tutor

Ann Arbor, MI

May 2023 - July 2023

OAKDALE NEIGHBORS

Tutor

Grand Rapids, MI

Sept. 2018 - Dec. 2019

Work Experience

UNIVERSITY OF MICHIGAN

Family Helpers

Orientation Leader - New Student Operations

Engineering Campus Tour Guide

Ann Arbor, MI

June 2023 - Aug. 2023

Feb. 2024 - May 2024

MICHIGAN MEDICINE RADIOLOGY DEPARTMENT

Patient Service Associate

Ann Arbor, MI

Jan. 2023 - April 2023

BAYCLIFF HEALTH CAMP

Counselor

Big Bay, MI

June 2022 - Aug. 2022

MICHIGAN DAILY

Opinion Cartoonist

Ann Arbor, MI

Sept. 2021 - May 2022

MEIJER GROCERY STORE

Cashier

Grand Rapids, MI

June 2021 - Aug. 2021, May 2022

BEACON HILL AT EASTGATE
Dining Server

Grand Rapids, MI
Oct. 2019 - May 2021

Leadership and Involvement

AMERICAN SIGN LANGUAGE CLUB
VP Communications
Vice President
Co-President

Ann Arbor, MI
Sept. 2024 - Dec. 2024
Sept. 2023 - May 2024
Jan. 2025 - May 2025

PHI SIGMA RHO PROFESSIONAL ENGINEERING SORORITY
Community Service Chair
Scholarship Chair

Ann Arbor, MI

PAGES FOR PEDIATRICS
Artist Team Member

Ann Arbor, MI

UNIVERSITY OF MICHIGAN CONCRETE CANOE TEAM
Team Member

Ann Arbor, MI

Professional Associations

ASSOCIATION FOR RESEARCH IN OTOLARYNGOLOGY

PHI SIGMA RHO PROFESSIONAL SORORITY

AMERICAN SOCIETY OF CIVIL ENGINEERS

Community Service

GROUNDCOVER NEWS
Volunteer

Ann Arbor, MI
May 2025 - Current

- Welcome and connect with the homeless or housing-insecure population of vendors who sell the newspaper
- Keep track of the transactions for selling the newspaper to vendors, 10-50 transactions per day

PURPLE HOUSE OVERNIGHT WARMING CENTER
Volunteer

Ann Arbor, MI
Dec. 2025

- Completed deescalation training

UNIVERSITY OF MICHIGAN
Michigan Medicine - Internal Medicine Unit Volunteer
Michigan Medicine - Sleep Center Volunteer
Campus Farm Work Days Volunteer

Ann Arbor, MI
Jan. 2026 - April 2026
Sept. 2025 - Dec. 2025
Sept. 2022, 2023, 2024

OAKDALE NEIGHBORS
Tutoring Volunteer

Grand Rapids, MI
Sept. 2018 - Dec. 2019

Scholarships and Awards

UNIVERSITY OF MICHIGAN ENGINEERING HONORS

2025

DAAD SCHOLARSHIP

2024

